

# Apply filters to SQL queries

Portfolio activity - SQL filters for a security investigation

## Project description

In this project, I used SQL filters to investigate potential security issues involving login attempts and employee machines. I queried the `log_in_attempts` and `employees` tables to retrieve records based on times, dates, countries, departments, and office locations. The goal was to narrow large datasets into records that were relevant to the investigation and to identify which employee machines needed security updates.

## Retrieve after-hours failed login attempts

**Scenario:** A potential security incident occurred after business hours, so I needed to review failed login attempts after 18:00.

### SQL query

```
SELECT *
FROM log_in_attempts
WHERE login_time > '18:00'
AND success = 0;
```

### How the query works

This query returns all records from `log_in_attempts` where the login attempt happened after 18:00 and was unsuccessful. The `login_time` condition filters for after-hours activity, and `success = 0` filters for failed attempts. I used `AND` because both conditions have to be true for a record to be included.

### Screenshot of query result

```
MariaDB [organization]> SELECT * FROM log_in_attempts WHERE login_time > '18:00' AND success = 0;
```

| event_id | username | login_date | login_time | country | ip_address      | success |
|----------|----------|------------|------------|---------|-----------------|---------|
| 2        | apatel   | 2022-05-10 | 20:27:27   | CAN     | 192.168.205.12  | 0       |
| 18       | pwashing | 2022-05-11 | 19:28:50   | US      | 192.168.66.142  | 0       |
| 20       | tshah    | 2022-05-12 | 18:56:36   | MEXICO  | 192.168.109.50  | 0       |
| 28       | astrada  | 2022-05-09 | 19:28:12   | MEXICO  | 192.168.27.57   | 0       |
| 34       | drosas   | 2022-05-11 | 21:02:04   | US      | 192.168.45.93   | 0       |
| 42       | cgriffin | 2022-05-09 | 23:04:05   | US      | 192.168.4.157   | 0       |
| 52       | cjackson | 2022-05-10 | 22:07:07   | CAN     | 192.168.58.57   | 0       |
| 69       | wjaffrey | 2022-05-11 | 19:55:15   | USA     | 192.168.100.17  | 0       |
| 82       | abernard | 2022-05-12 | 23:38:46   | MEX     | 192.168.234.49  | 0       |
| 87       | apatel   | 2022-05-08 | 22:38:31   | CANADA  | 192.168.132.153 | 0       |
| 96       | ivelasco | 2022-05-09 | 22:36:36   | CAN     | 192.168.84.194  | 0       |
| 104      | asundara | 2022-05-11 | 18:38:07   | US      | 192.168.96.200  | 0       |
| 107      | bisles   | 2022-05-12 | 20:25:57   | USA     | 192.168.116.187 | 0       |
| 111      | astrada  | 2022-05-10 | 22:00:26   | MEXICO  | 192.168.76.27   | 0       |
| 127      | abellmas | 2022-05-09 | 21:20:51   | CANADA  | 192.168.70.122  | 0       |
| 131      | bisles   | 2022-05-09 | 20:03:55   | US      | 192.168.113.171 | 0       |
| 155      | cgriffin | 2022-05-12 | 22:18:42   | USA     | 192.168.236.176 | 0       |
| 160      | jclark   | 2022-05-10 | 20:49:00   | CANADA  | 192.168.214.49  | 0       |
| 199      | yappiah  | 2022-05-11 | 19:34:48   | MEXICO  | 192.168.44.232  | 0       |

```
19 rows in set (0.001 sec)
```

Figure 1. Failed login attempts after 18:00.

## Retrieve login attempts on specific dates

**Scenario:** A suspicious event occurred on 2022-05-09, so I reviewed login attempts from that date and the previous day.

### SQL query

```
SELECT *
FROM log_in_attempts
WHERE login_date = '2022-05-09'
OR login_date = '2022-05-08';
```

### How the query works

This query returns login attempts from either 2022-05-09 or 2022-05-08. The OR operator is used because a record only needs to match one of the two dates to be returned. This made it possible to review activity from the suspicious date and the day before it.

### Screenshot of query result

MariaDB [organization]> SELECT \* FROM log\_in\_attempts WHERE login\_date = '2022-05-09' OR login\_date = '2022-05-08';

| event_id | username | login_date | login_time | country | ip_address      | success |
|----------|----------|------------|------------|---------|-----------------|---------|
| 1        | jrafael  | 2022-05-09 | 04:56:27   | CAN     | 192.168.243.140 | 1       |
| 3        | dkot     | 2022-05-09 | 06:47:41   | USA     | 192.168.151.162 | 1       |
| 4        | dkot     | 2022-05-08 | 02:00:39   | USA     | 192.168.178.71  | 0       |
| 8        | bisles   | 2022-05-08 | 01:30:17   | US      | 192.168.119.173 | 0       |
| 12       | dkot     | 2022-05-08 | 09:11:34   | USA     | 192.168.100.158 | 1       |
| 15       | lyamamot | 2022-05-09 | 17:17:26   | USA     | 192.168.183.51  | 0       |
| 24       | arusso   | 2022-05-09 | 06:49:39   | MEXICO  | 192.168.171.192 | 1       |
| 25       | sbaelish | 2022-05-09 | 07:04:02   | US      | 192.168.33.137  | 1       |
| 26       | apatel   | 2022-05-08 | 17:27:00   | CANADA  | 192.168.123.105 | 1       |
| 28       | astrada  | 2022-05-09 | 19:28:12   | MEXICO  | 192.168.27.57   | 0       |
| 30       | yappiah  | 2022-05-09 | 03:22:22   | MEX     | 192.168.124.48  | 1       |
| 32       | acook    | 2022-05-09 | 02:52:02   | CANADA  | 192.168.142.239 | 0       |
| 36       | asundara | 2022-05-08 | 09:00:42   | US      | 192.168.78.151  | 1       |
| 38       | sbaelish | 2022-05-09 | 14:40:01   | USA     | 192.168.60.42   | 1       |
| 39       | yappiah  | 2022-05-09 | 07:56:40   | MEXICO  | 192.168.57.115  | 1       |
| 42       | cgriffin | 2022-05-09 | 23:04:05   | US      | 192.168.4.157   | 0       |
| 43       | mcouliba | 2022-05-08 | 02:35:34   | CANADA  | 192.168.16.208  | 0       |
| 44       | daquino  | 2022-05-08 | 07:02:35   | CANADA  | 192.168.168.144 | 0       |
| 47       | dkot     | 2022-05-08 | 05:06:45   | US      | 192.168.233.24  | 1       |
| 49       | asundara | 2022-05-08 | 14:00:01   | US      | 192.168.173.213 | 0       |
| 53       | nmason   | 2022-05-08 | 11:51:38   | CAN     | 192.168.133.188 | 1       |
| 56       | acook    | 2022-05-08 | 04:56:30   | CAN     | 192.168.209.130 | 1       |
| 58       | ivelasco | 2022-05-09 | 17:20:54   | CAN     | 192.168.57.162  | 0       |
| 61       | dtanaka  | 2022-05-09 | 09:45:18   | USA     | 192.168.98.221  | 1       |
| 65       | aalonso  | 2022-05-09 | 23:42:12   | MEX     | 192.168.52.37   | 1       |
| 66       | astrada  | 2022-05-08 | 21:58:32   | MEX     | 192.168.67.223  | 1       |
| 67       | abernard | 2022-05-09 | 11:53:41   | MEX     | 192.168.118.29  | 1       |
| 68       | mrah     | 2022-05-08 | 17:16:13   | US      | 192.168.42.248  | 1       |
| 70       | tmitchel | 2022-05-09 | 10:55:17   | MEXICO  | 192.168.87.199  | 1       |
| 71       | mcouliba | 2022-05-09 | 06:57:42   | CAN     | 192.168.55.169  | 0       |
| 72       | alevitsk | 2022-05-08 | 12:09:10   | CANADA  | 192.168.139.176 | 1       |
| 79       | abernard | 2022-05-09 | 11:41:15   | MEX     | 192.168.158.170 | 0       |
| 80       | cjackson | 2022-05-08 | 02:18:10   | CANADA  | 192.168.33.140  | 1       |
| 82       | lradgiew | 2022-05-08 | 08:10:22   | USA     | 192.168.57.58   | 1       |

Figure 2. Login attempts from 2022-05-09 and 2022-05-08.

## Retrieve login attempts outside of Mexico

**Scenario:** The team determined that the suspicious activity did not originate in Mexico, so I filtered out Mexico-related records.

### SQL query

```
SELECT *
FROM log_in_attempts
WHERE country NOT LIKE 'MEX%';
```

### How the query works

This query returns login attempts from countries outside of Mexico. The pattern LIKE 'MEX%' matches country values that begin with MEX, including MEX and MEXICO. Adding NOT excludes those matching values and returns records from other countries.

### Screenshot of query result

```
MariaDB [organization]> SELECT * FROM log_in_attempts WHERE country NOT LIKE 'MEX%';
```

| event_id | username | login_date | login_time | country | ip_address      | success |
|----------|----------|------------|------------|---------|-----------------|---------|
| 1        | jrafael  | 2022-05-09 | 04:56:27   | CAN     | 192.168.243.140 | 1       |
| 2        | apatel   | 2022-05-10 | 20:27:27   | CAN     | 192.168.205.12  | 0       |
| 3        | dkot     | 2022-05-09 | 06:47:41   | USA     | 192.168.151.162 | 1       |
| 4        | dkot     | 2022-05-08 | 02:00:39   | USA     | 192.168.178.71  | 0       |
| 5        | jrafael  | 2022-05-11 | 03:05:59   | CANADA  | 192.168.86.232  | 0       |
| 7        | eraab    | 2022-05-11 | 01:45:14   | CAN     | 192.168.170.243 | 1       |
| 8        | bisles   | 2022-05-08 | 01:30:17   | US      | 192.168.119.173 | 0       |
| 10       | jrafael  | 2022-05-12 | 09:33:19   | CANADA  | 192.168.228.221 | 0       |
| 11       | sgilmore | 2022-05-11 | 10:16:29   | CANADA  | 192.168.140.81  | 0       |
| 12       | dkot     | 2022-05-08 | 09:11:34   | USA     | 192.168.100.158 | 1       |
| 13       | mrah     | 2022-05-11 | 09:29:34   | USA     | 192.168.246.135 | 1       |
| 14       | sbaelish | 2022-05-10 | 10:20:18   | US      | 192.168.16.99   | 1       |
| 15       | lyamamot | 2022-05-09 | 17:17:26   | USA     | 192.168.183.51  | 0       |
| 16       | mcouliba | 2022-05-11 | 06:44:22   | CAN     | 192.168.172.189 | 1       |
| 17       | pwashing | 2022-05-11 | 02:33:02   | USA     | 192.168.81.89   | 1       |
| 18       | pwashing | 2022-05-11 | 19:28:50   | US      | 192.168.66.142  | 0       |
| 19       | jhill    | 2022-05-12 | 13:09:04   | US      | 192.168.142.245 | 1       |
| 21       | iuduike  | 2022-05-11 | 17:50:00   | US      | 192.168.131.147 | 1       |
| 25       | sbaelish | 2022-05-09 | 07:04:02   | US      | 192.168.33.137  | 1       |
| 26       | apatel   | 2022-05-08 | 17:27:00   | CANADA  | 192.168.123.105 | 1       |
| 29       | bisles   | 2022-05-11 | 01:21:22   | US      | 192.168.85.186  | 0       |
| 31       | acook    | 2022-05-12 | 17:36:45   | CANADA  | 192.168.58.232  | 0       |
| 32       | acook    | 2022-05-09 | 02:52:02   | CANADA  | 192.168.142.239 | 0       |
| 33       | zbernal  | 2022-05-11 | 02:52:10   | US      | 192.168.72.59   | 1       |
| 34       | drosas   | 2022-05-11 | 21:02:04   | US      | 192.168.45.93   | 0       |
| 36       | asundara | 2022-05-08 | 09:00:42   | US      | 192.168.78.151  | 1       |
| 37       | eraab    | 2022-05-10 | 06:03:41   | CANADA  | 192.168.152.148 | 0       |
| 38       | sbaelish | 2022-05-09 | 14:40:01   | USA     | 192.168.60.42   | 1       |
| 41       | apatel   | 2022-05-10 | 17:39:42   | CANADA  | 192.168.46.207  | 0       |
| 42       | cgriffin | 2022-05-09 | 23:04:05   | US      | 192.168.4.157   | 0       |
| 43       | mcouliba | 2022-05-08 | 02:35:34   | CANADA  | 192.168.16.208  | 0       |
| 44       | daquino  | 2022-05-08 | 07:02:35   | CANADA  | 192.168.168.144 | 0       |
| 45       | dtanaka  | 2022-05-11 | 10:28:54   | US      | 192.168.223.157 | 1       |
| 46       | eraab    | 2022-05-11 | 11:29:27   | CAN     | 192.168.24.12   | 0       |

Figure 3. Login attempts that did not originate from Mexico.

## Retrieve Marketing employees in the East building

**Scenario:** The team needed to perform security updates on machines for Marketing employees located in East building offices.

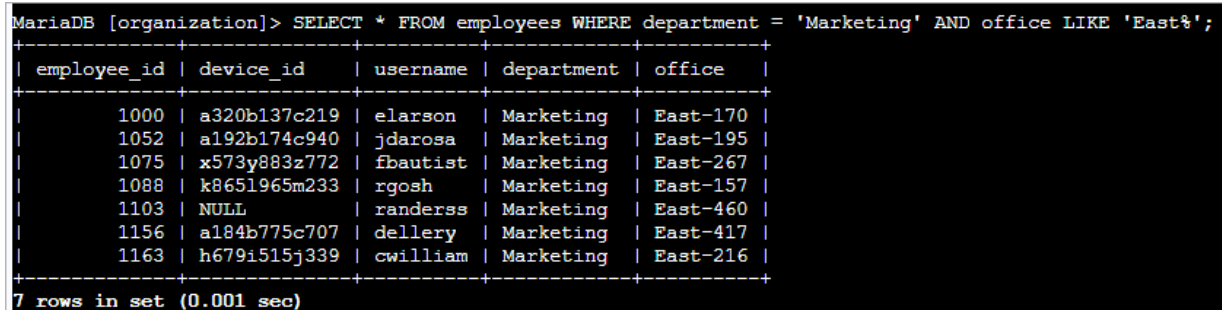
### SQL query

```
SELECT *  
FROM employees  
WHERE department = 'Marketing'  
AND office LIKE 'East%';
```

### How the query works

This query returns employees who are in the Marketing department and have an office value that starts with East. The LIKE 'East%' pattern matches offices such as East-170, East-267, and East-460. I used AND because both the department and office conditions must be true.

### Screenshot of query result



```
MariaDB [organization]> SELECT * FROM employees WHERE department = 'Marketing' AND office LIKE 'East%';  
+-----+-----+-----+-----+-----+  
| employee_id | device_id | username | department | office |  
+-----+-----+-----+-----+-----+  
| 1000 | a320b137c219 | elarson | Marketing | East-170 |  
| 1052 | a192b174c940 | jdarosa | Marketing | East-195 |  
| 1075 | x573y883z772 | fbautist | Marketing | East-267 |  
| 1088 | k865l965m233 | rgosh | Marketing | East-157 |  
| 1103 | NULL | randeress | Marketing | East-460 |  
| 1156 | a184b775c707 | dellery | Marketing | East-417 |  
| 1163 | h679i515j339 | cwilliam | Marketing | East-216 |  
+-----+-----+-----+-----+-----+  
7 rows in set (0.001 sec)
```

Figure 4. Marketing employees located in East building offices.

## Retrieve employees in Finance or Sales

**Scenario:** The team needed to identify machines for employees in either the Finance or Sales departments.

### SQL query

```
SELECT *
FROM employees
WHERE department = 'Finance'
OR department = 'Sales';
```

### How the query works

This query returns employees who work in Finance or Sales. I used OR because an employee only needs to belong to one of those departments to be included in the results. This helps identify the employee machines that need the security update for these two departments.

### Screenshot of query result

MariaDB [organization]> SELECT \* FROM employees WHERE department = 'Finance' OR department = 'Sales';

| employee_id | device_id    | username | department | office      |
|-------------|--------------|----------|------------|-------------|
| 1003        | d394e816f943 | sgilmore | Finance    | South-153   |
| 1007        | h174i497j413 | wjaffrey | Finance    | North-406   |
| 1008        | i858j583k571 | abernard | Finance    | South-170   |
| 1009        | NULL         | lrodriqu | Sales      | South-134   |
| 1010        | k242l212m542 | jlansky  | Finance    | South-109   |
| 1011        | l748m120n401 | drosas   | Sales      | South-292   |
| 1015        | p611q262r945 | jsoto    | Finance    | North-271   |
| 1017        | r550s824t230 | jclark   | Finance    | North-188   |
| 1018        | s310t540u653 | abellmas | Finance    | North-403   |
| 1022        | w237x430y567 | arusso   | Finance    | West-465    |
| 1024        | y976z753a267 | iuduike  | Sales      | South-215   |
| 1025        | z381a365b233 | jhill    | Sales      | North-115   |
| 1029        | d336e475f676 | ivelasco | Finance    | East-156    |
| 1035        | j236k303l245 | bisles   | Sales      | South-171   |
| 1039        | n253o917p623 | cjackson | Sales      | East-378    |
| 1041        | p929q222r778 | cgriffin | Sales      | North-208   |
| 1044        | s429t157u159 | tbarnes  | Finance    | West-415    |
| 1045        | t567u844v434 | pwashing | Finance    | East-115    |
| 1046        | u429v921w138 | daquino  | Finance    | West-280    |
| 1047        | v109w587x644 | cward    | Finance    | West-373    |
| 1048        | w167x592y375 | tmitchel | Finance    | South-288   |
| 1049        | NULL         | jreckley | Finance    | Central-295 |
| 1050        | y132z930a114 | csimmons | Finance    | North-468   |
| 1057        | f370g535h632 | mscott   | Sales      | South-270   |
| 1062        | k367l639m697 | redwards | Finance    | North-180   |
| 1063        | l686m140n569 | lpope    | Sales      | East-226    |
| 1066        | o678p794q957 | ttyrell  | Sales      | Central-444 |

Figure 5. Employees in Finance or Sales.

## Retrieve all employees not in Information Technology

**Scenario:** Employees in the Information Technology department already received an update, so I needed to find employees in all other departments.

### SQL query

```
SELECT *
FROM employees
WHERE department NOT LIKE 'Information Technology';
```

### How the query works

This query returns employees whose department is not Information Technology. Since there is no wildcard in the pattern, NOT LIKE works like a not-equal comparison in this case. I used NOT to exclude Information Technology employees and return the remaining departments that still needed the update.

### Screenshot of query result

```
MariaDB [organization]> SELECT * FROM employees WHERE department NOT LIKE 'Information Technology';
```

| employee_id | device_id    | username | department      | office      |
|-------------|--------------|----------|-----------------|-------------|
| 1000        | a320b137c219 | elarson  | Marketing       | East-170    |
| 1001        | b239c825d303 | bmoreno  | Marketing       | Central-276 |
| 1002        | c116d593e558 | tshah    | Human Resources | North-434   |
| 1003        | d394e816f943 | sgilmore | Finance         | South-153   |
| 1004        | e218f877g788 | eraab    | Human Resources | South-127   |
| 1005        | f551g340h864 | gesparza | Human Resources | South-366   |
| 1007        | h174i497j413 | wjaffrey | Finance         | North-406   |
| 1008        | i858j583k571 | abernard | Finance         | South-170   |
| 1009        | NULL         | lrodriqu | Sales           | South-134   |
| 1010        | k242l212m542 | jlansky  | Finance         | South-109   |
| 1011        | l748m120n401 | drosas   | Sales           | South-292   |
| 1015        | p611q262r945 | jsoto    | Finance         | North-271   |
| 1016        | q793r736s288 | sbaelish | Human Resources | North-229   |
| 1017        | r550s824t230 | jclark   | Finance         | North-188   |
| 1018        | s310t540u653 | abellmas | Finance         | North-403   |
| 1020        | u899v381w363 | arutley  | Marketing       | South-351   |
| 1022        | w237x430y567 | arusso   | Finance         | West-465    |
| 1024        | y976z753a267 | iuduike  | Sales           | South-215   |
| 1025        | z381a365b233 | jhill    | Sales           | North-115   |
| 1026        | a998b568c863 | apatel   | Human Resources | West-320    |
| 1027        | b806c503d354 | mrah     | Marketing       | West-246    |
| 1028        | c603d749e374 | aestrada | Human Resources | West-121    |
| 1029        | d336e475f676 | ivelasco | Finance         | East-156    |
| 1030        | e391f189g913 | mabadi   | Marketing       | West-375    |
| 1031        | f419g188h578 | dkot     | Marketing       | West-408    |
| 1034        | i679j565k940 | bsand    | Human Resources | East-484    |
| 1035        | j236k303l245 | bisles   | Sales           | South-171   |
| 1036        | k550l533m205 | rjensen  | Marketing       | Central-239 |

Figure 6. Employees who are not in Information Technology.

## Summary

I used SQL queries to filter login and employee data for a security investigation. The queries used AND to combine required conditions, OR to search for multiple possible values, and NOT to exclude records that were not relevant to the investigation. I also used LIKE with the % wildcard to search for patterns in country and office values. These filters helped retrieve targeted information from larger datasets and supported decisions about suspicious login activity and employee machine updates.

## Portfolio requirements covered

- ✓ Screenshots or typed versions of the SQL queries
- ✓ Project description at the beginning
- ✓ Explanation of each query and how it works
- ✓ Summary at the end
- ✓ Use of LIKE with % to search for patterns
- ✓ Filtering by dates and times
- ✓ Use of AND and OR to combine conditions
- ✓ Use of NOT to exclude records